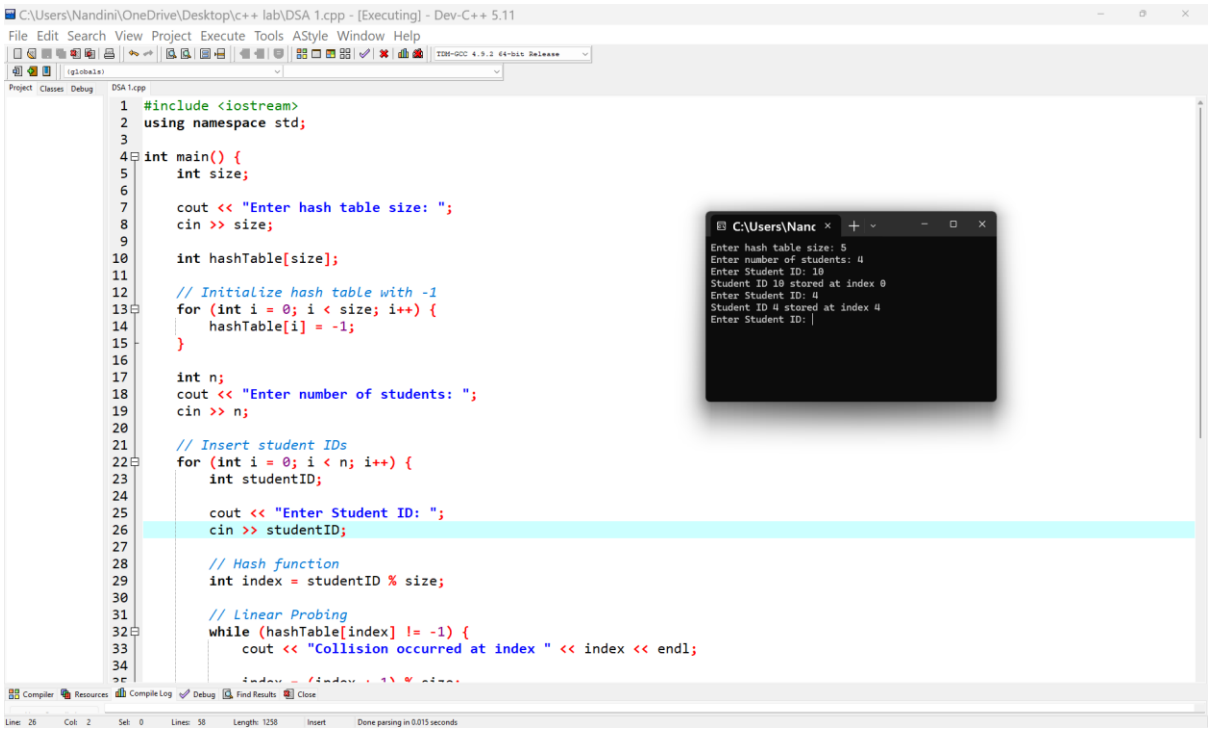
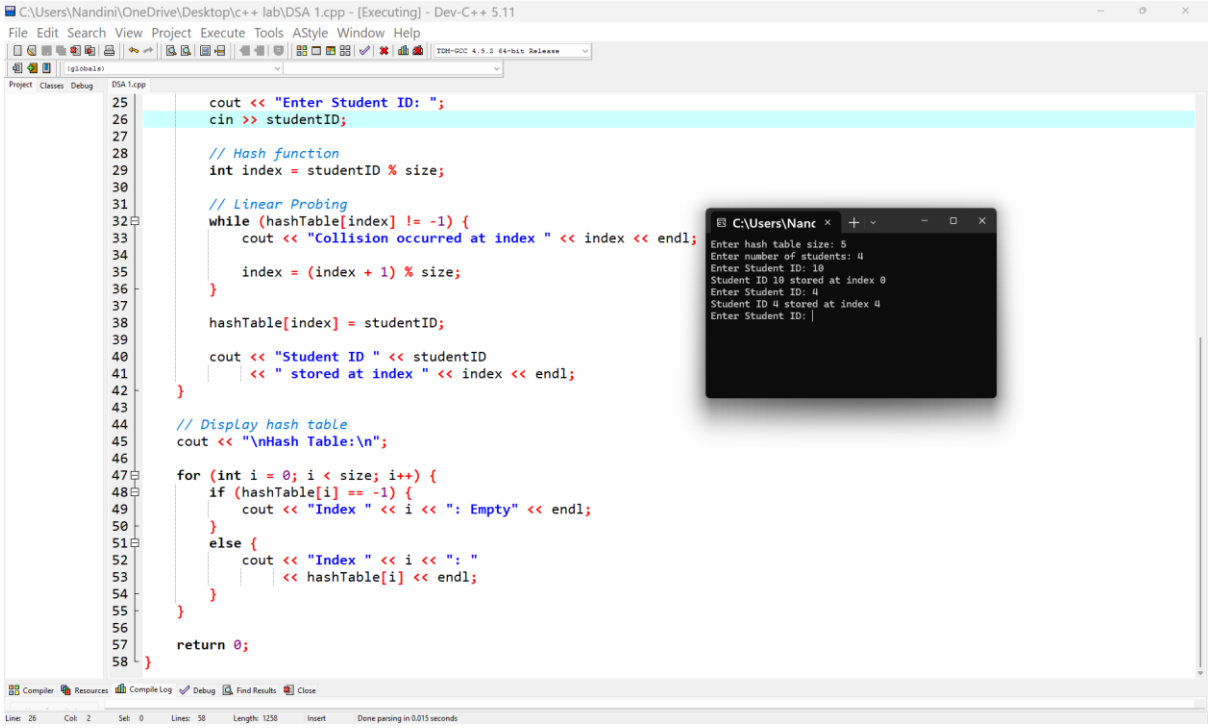
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Batch: AI&ML(B1)

Student ID Management using Linear Probing





CODE:

```
#include <iostream>
using namespace std;

int main() {
   int size;

   cout << "Enter hash table size: ";
   cin >> size;

   int hashTable[size];

   // Initialize hash table with -1
   for (int i = 0; i < size; i++) {
      hashTable[i] = -1;
   }

   int n;
   cout << "Enter number of students: ";
   cin >> n;

   // Insert student IDs
   for (int i = 0; i < n; i++) {
      int studentID;

      cout << "Enter Student ID: ";
      cin >> studentID;
```

```
    // Hash function
    int index = studentID % size;

    // Linear Probing
    while (hashTable[index] != -1) {
        cout << "Collision occurred at index " << index << endl;

        index = (index + 1) % size;
    }

    hashTable[index] = studentID;

    cout << "Student ID " << studentID
         << " stored at index " << index << endl;
}

// Display hash table
cout << "\nHash Table:\n";

for (int i = 0; i < size; i++) {
    if (hashTable[i] == -1) {
        cout << "Index " << i << ": Empty" << endl;
    }
    else {
        cout << "Index " << i << ": "
             << hashTable[i] << endl;
    }
}
```

```
    return 0;
}
```